

COL AT3

SIMATS ENGINEERING
ASSESSMENT TOOLS

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

CLASS TEST

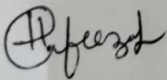
Code of Conduct:

Shaik Hafeeza Tarannum

I 192372044 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



SIMATS ENGINEERING ASSESSMENT TOOLS

1. Design a PDA to accept strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b', where 'n' is a non-negative integer.

$$i) L = \{a^n b^n\}$$

2. Convert the given context-free grammar (CFG) into a Pushdown Automaton (PDA) that recognizes the same language.

$$E \rightarrow E+E \mid E^*E \mid (E) \mid 1$$

$$I \rightarrow Ia \mid Ib \mid I0 \mid I1 \mid a \mid b$$

3. Design a TM to accept the strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b', where 'n' is a non-negative integer

$$i) L = \{0^n 1^n \mid n \geq 1\}$$

4. Design a Pushdown Automata for the language representing a's followed by b's and the number of b's must be twice that of a's.

$$L = \{a^n b^{2n} \mid n \geq 1\}$$

5. Construct a Turing Machine to recognize the language. Write the logic used for the design and the formal definition

$$L = \{a^n b^n c^n \mid n \geq 1\}$$

CLASSTEST-2Given $L = \{a^n b^n\}$

To design a PDA to accept the strings "with 'n' occurrences of 'a' and followed by 'n' occurrences of 'b'."
 where n is non-negative integer.

Idea:

- ⇒ For every 'a', push one symbol A onto the stack
- ⇒ For every 'b' pop one A
- ⇒ Accept when all the input is read and only the bottom stack symbol remains.

PDA Definition:

$$M = \{Q, \Sigma, \Gamma, \delta, S, F, q_0\}$$

where $Q = \{q_0, q_1, q_2\}$ [set of finite states] $\Sigma = \{a, b\}$ [Alphabet] $\Gamma = A, Z_0$ $S = q_0$ [Start state] $F = q_f$ [Final state] $q_0 = Z_0$ [Initial stack symbol]Transitions: δ = Transitions

$$\delta(q_0, a, Z_0) = (q_0, AZ_0)$$

$$\delta(q_0, a, A) = (q_0, AA)$$

$$\delta(q_0, b, A) = (q_1, \epsilon)$$

$$\delta(q_1, b, A) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, Z_0) = (q_f, Z_0)$$

$$\delta(q_1, \epsilon, Z_0) = (q_f, Z_0)$$

example:

For input aaabbb:

$$q_0 \xrightarrow{aaa} \text{push AAA} \xrightarrow{bbb} \text{pop AAA} \rightarrow q_f$$

For I

$$\delta(q_1, \epsilon, I) = (q_1, I_a)$$

$$\delta(q_1, \epsilon, I) = (q_1, I_b)$$

$$\delta(q_1, \epsilon, I) = (q_1, I_0)$$

$$\delta(q_1, \epsilon, I) = (q_1, I_1)$$

$$\delta(q_1, \epsilon, I) = (q_1, a)$$

$$\delta(q_1, \epsilon, I) = (q_1, b)$$

Step 3: match terminals:

$$\delta(q_1, a, a) \rightarrow \epsilon$$

$$\delta(q_1, b, b) \rightarrow \epsilon$$

$$\delta(q_1, 0, 0) \rightarrow \epsilon$$

$$\delta(q_1, 1, 1) \rightarrow \epsilon$$

$$\delta(q_1, *, *) \rightarrow \epsilon$$

$$\delta(q_1, *, *) \rightarrow \epsilon$$

$$\delta(q_1, \epsilon, \epsilon) \rightarrow \epsilon$$

$$\delta(q_1, \epsilon, \epsilon) = \epsilon$$

Step 4: Accept

$$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$$

The PDA simulates the CFG by replacing variables using productions and matching terminals with the input

Turing Machine for

$$L = \{0^n 1^n \mid n \geq 1\}$$

$\Rightarrow$ Find the leftmost unmarked 0 and replace it with x

$\Rightarrow$ Move right and find first unmarked 1; replace it with y.

$\Rightarrow$ Return to left

$\Rightarrow$ Repeat until all 0s and 1s are marked

$\Rightarrow$ Accept if the number of 0s and 1s are equal

$$M = \{Q, \Sigma, \Gamma, q_0, \delta, F\}$$

Where $Q = \{q_0, q_1, q_2, q_3, q_{accept}, q_{reject}\}$

$$\Sigma = \{0, 1\}$$

$$\Gamma = \{0, 1, x, y, \epsilon\}$$

$$F = \{q_{accept}\}$$

Transitions:

$$\delta(q_0, \epsilon) = (q_1, X, R)$$

$$\delta(q_0, x) = (q_0, x, R)$$

$$\delta(q_1, \epsilon) = (q_1, \epsilon, R)$$

$$\delta(q_1, y) = (q_1, y, R)$$

$$\delta(q_1, l) = (q_2, y, L)$$

Return left:

$$\delta(q_2, \epsilon) = (q_2, \epsilon, L)$$

$$\delta(q_2, l) = (q_2, l, L)$$

$$\delta(q_2, x) = (q_0, x, R)$$

Accept: when all symbols are marked

$$\delta(q_0, y) = (q_3, y, R)$$

$$\delta(q_3, y) = (q_3, y, R)$$

$$\delta(q_3, B) = (q_{\text{accept}}, B, R)$$

Ex:

0011.

$$0011 \rightarrow x0y1 \rightarrow xxyy$$

$\therefore$ The string is accepted

(4) POA for $L = \{a^n b^{2n} \mid n \geq 1\}$

$\Rightarrow$ For every a , push two stack symbols

$\Rightarrow$ For every b , pop one stack symbol

$\Rightarrow$ Therefore if there are n occurrences of a the stack contains 2^n symbols and exactly 2^n b s are required to empty it

$$M = \{Q, \Sigma, \delta, \Gamma, q_0, z_0, F\}$$

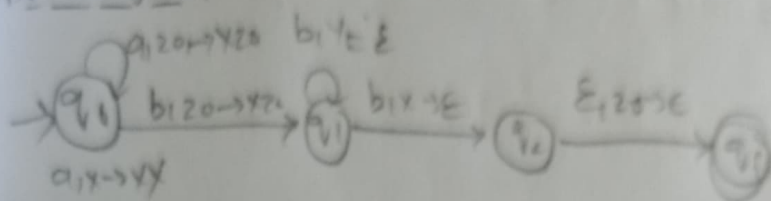
where $Q_0 = \{q_0, q_1, q_2\}$

$$\Sigma = \{a, b\}$$

$$\Gamma = A, z_0$$

$$F = \{q_f\}$$

ppa diagram:



Transitions

Current	Input	Stack	Action	Next
q0	a	z0	AAz0	z0
q0	a	A	AAA	q0
q0	b	A	ϵ	q1
q1	b	A	ϵ	q1
q1	ϵ	z0	ϵ	q3

Example

$$a^2b^4 = a^2abbbb$$

stack operation:

a $\rightarrow$ AA

a $\rightarrow$ AAA

b $\rightarrow$ pop A

b $\rightarrow$ pop A

b $\rightarrow$ pop A

b $\rightarrow$ pop A

stack $\rightarrow$ z0

Accepted.

5) TM for $L = \{a^n b^m c^n \mid n \geq 1\}$

For every a:

$\Rightarrow$ change a to x

$\Rightarrow$ Find one b and change it to y

$\Rightarrow$ Find one c and change it to z

$\Rightarrow$ Return to left

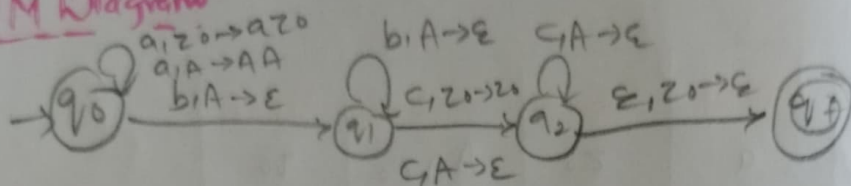
$\Rightarrow$ Repeat

$\Rightarrow$ If all a, b and c are matched $\Rightarrow$ accept

TM

$M = \{Q, \Sigma, \Gamma, \delta, q_0, z_0, F\}$

TM Diagram:



Important transitions

q_0

$a \rightarrow x, R$

$x \rightarrow x, R$

q_1

$a \rightarrow a, R$

$x \rightarrow x, R$

$b \rightarrow y, R$

q_2

$b \rightarrow b, R$

$y \rightarrow y, R$

$\epsilon \rightarrow z, L$

q_f :

$a \rightarrow a, L$

$b \rightarrow b, L$

$c \rightarrow c, L$

$x \rightarrow x, L$

$y \rightarrow y, L$

$z \rightarrow z, L$

$B \rightarrow B, R$

Example

Input

aaabbbccc (First round)

$\downarrow$

xaabybbzcc (Second round)

$\downarrow$

xxaxxbzzc

$\downarrow$

xxxxyyzzz

(Third round)

Accepted.

$\therefore$ The string is accepted.

SIMATS ENGINEERING ASSESSMENT TOOLS

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2-3 Marks)	Level 1 – Beginning (0-1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

94/100